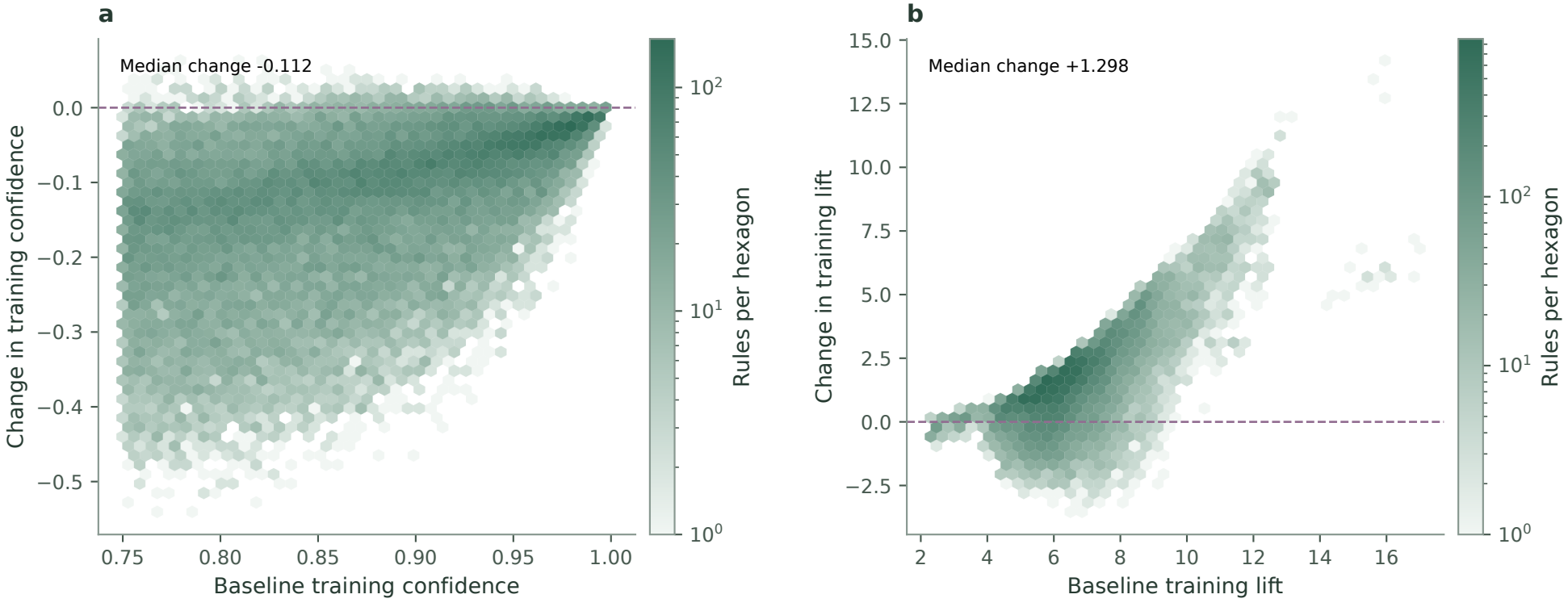



Fixed baseline rule family: n = 27,037; 16.6% remain above all discovery thresholds.
Removed broadest 5% of training species (n = 856); cluster ARI = 0.731. Dashed line: no metric change.